

Nama : Johannes Bagus Pramindra

Nim : 662023002

Username : bagusprime

```
1  [x, y] = meshgrid(-6:0.1:6, -6:0.1:6);
2  f = (x.^2 + y - 11).^2 + (x + y.^2 - 7).^2;
3
4  figure; surf(x, y, f); title('Himmelblau's Function'); shading interp;
5  figure; contour(x, y, f, 50); title('Contour of Himmelblau's');
6
7  [x, y] = meshgrid(-5.12:0.1:5.12, -5.12:0.1:5.12);
8  f = 20 + x.^2 - 10*cos(2*pi*x) + y.^2 - 10*cos(2*pi*y);
9
10 figure; surf(x, y, f); title('Rastrigin Function'); shading interp;
11 figure; contour(x, y, f, 50); title('Contour of Rastrigin');
12
13 [x, y] = meshgrid(-5:0.1:5, -5:0.1:5);
14 f = -20 * exp(-0.2*sqrt(0.5*(x.^2 + y.^2))) - ...
15     exp(0.5*(cos(2*pi*x) + cos(2*pi*y))) + exp(1) + 20;
16
17 figure; surf(x, y, f); title('Ackley Function'); shading interp;
18 figure; contour(x, y, f, 50); title('Contour of Ackley');
19
20 [x, y] = meshgrid(-10:0.1:10, -10:0.1:10);
21 f = (x + 2*y - 7).^2 + (2*x + y - 5).^2;
22
23 figure; surf(x, y, f); title('Booth Function'); shading interp;
24 figure; contour(x, y, f, 50); title('Contour of Booth');
25
```